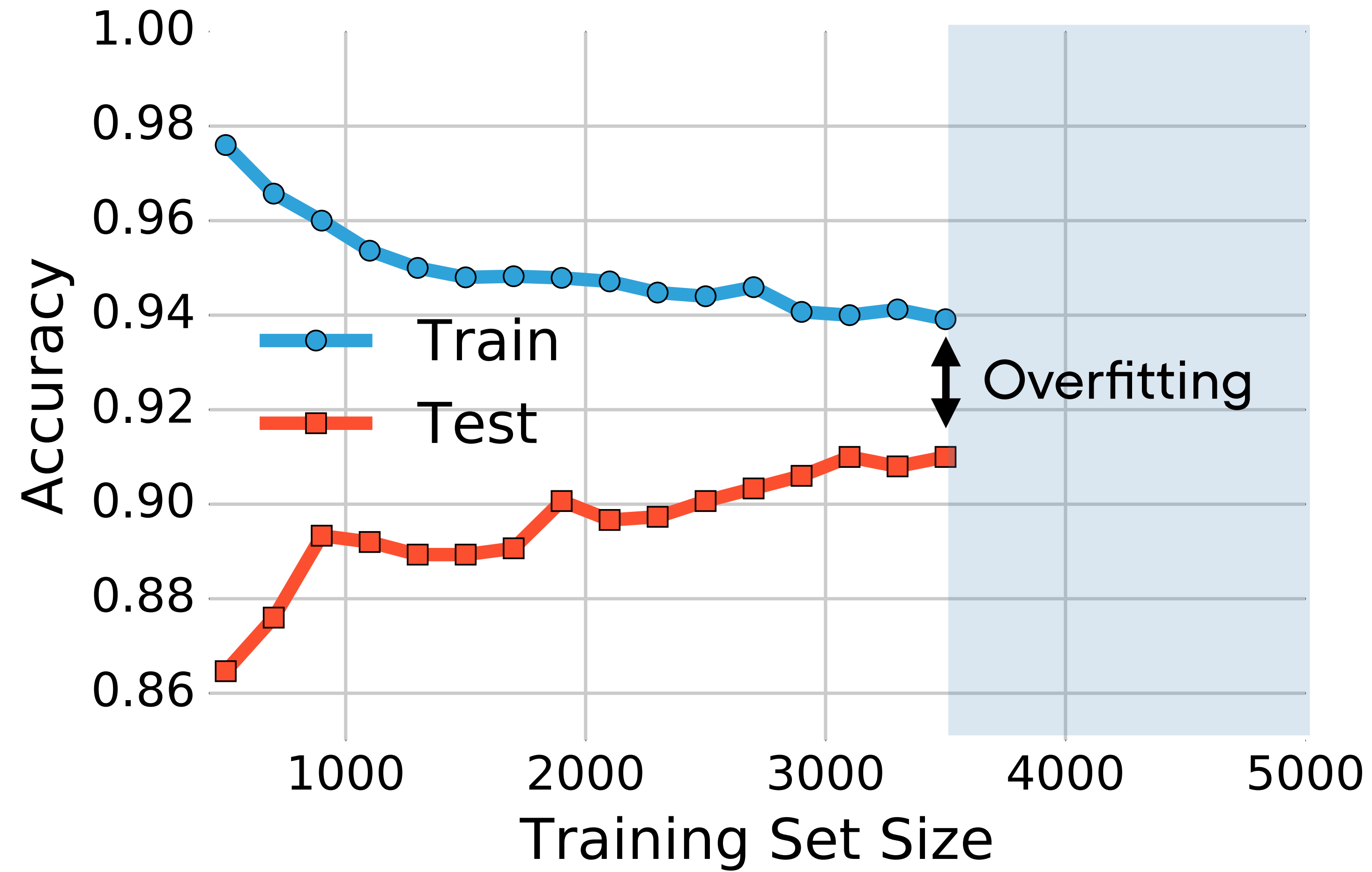


6.7

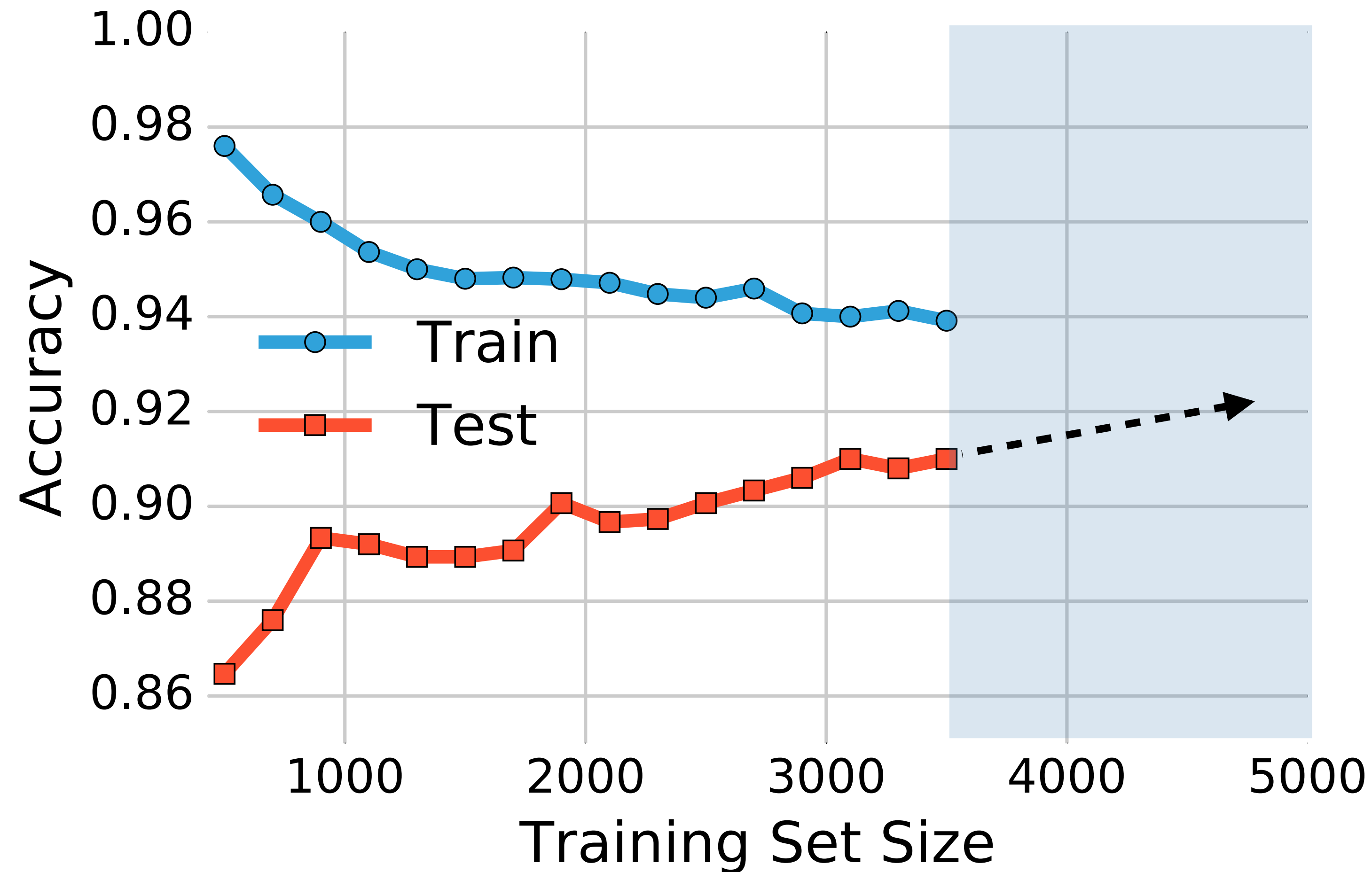
Reducing Overfitting With Dropout

Part 1: The Main Idea Behind Dropout

Sebastian Raschka and the Lightning AI Team



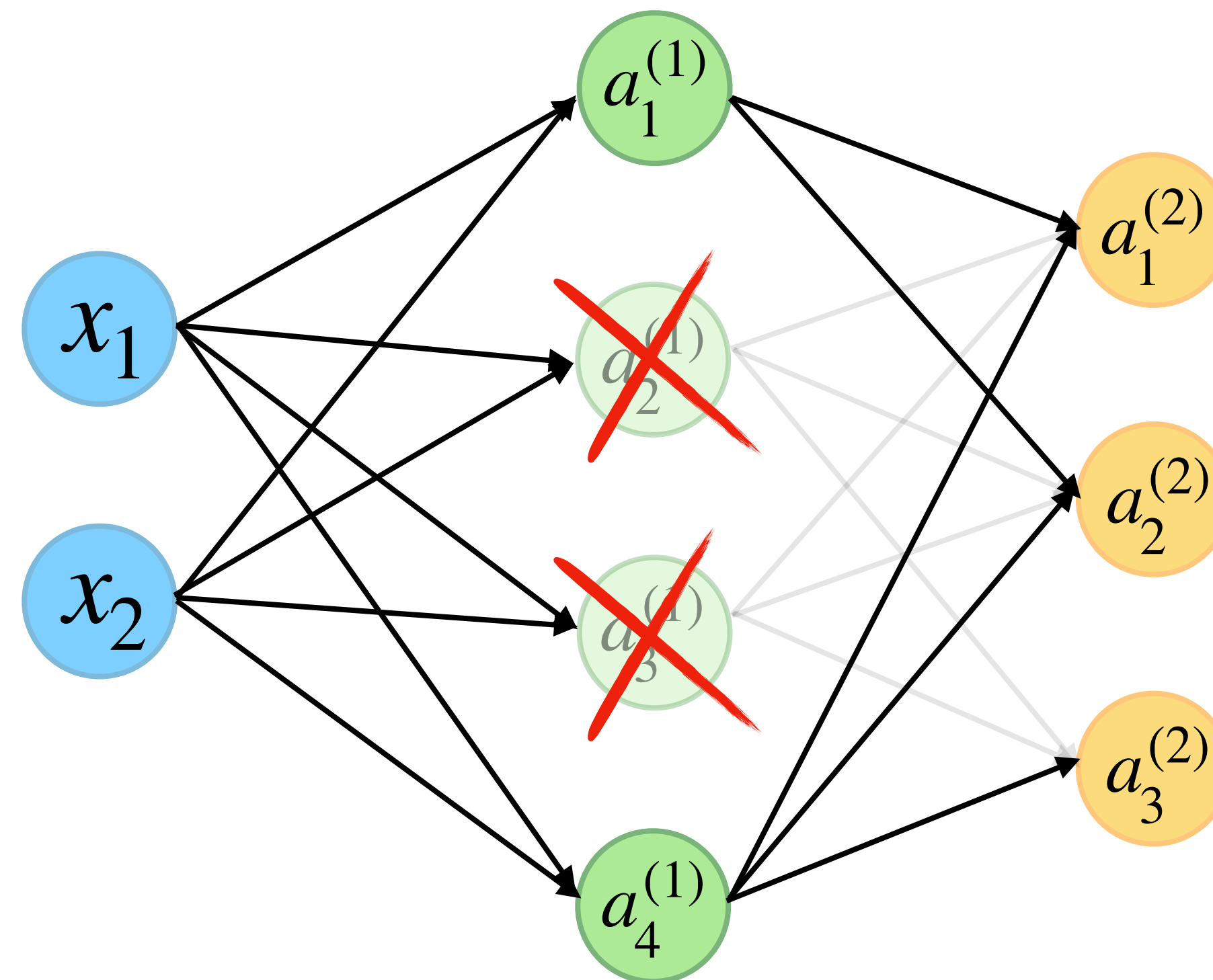
Often, **collecting more data** yields the "best bang for the buck"



Reducing Overfitting With Dropout

Hinton, Srivastava, Krizhevsky, Sutskever, Salakhutdinov (2012). Improving neural networks by preventing co-adaptation of feature detectors.
<https://arxiv.org/abs/1207.0580>

Dropout in a nutshell: **randomly drop nodes** during training

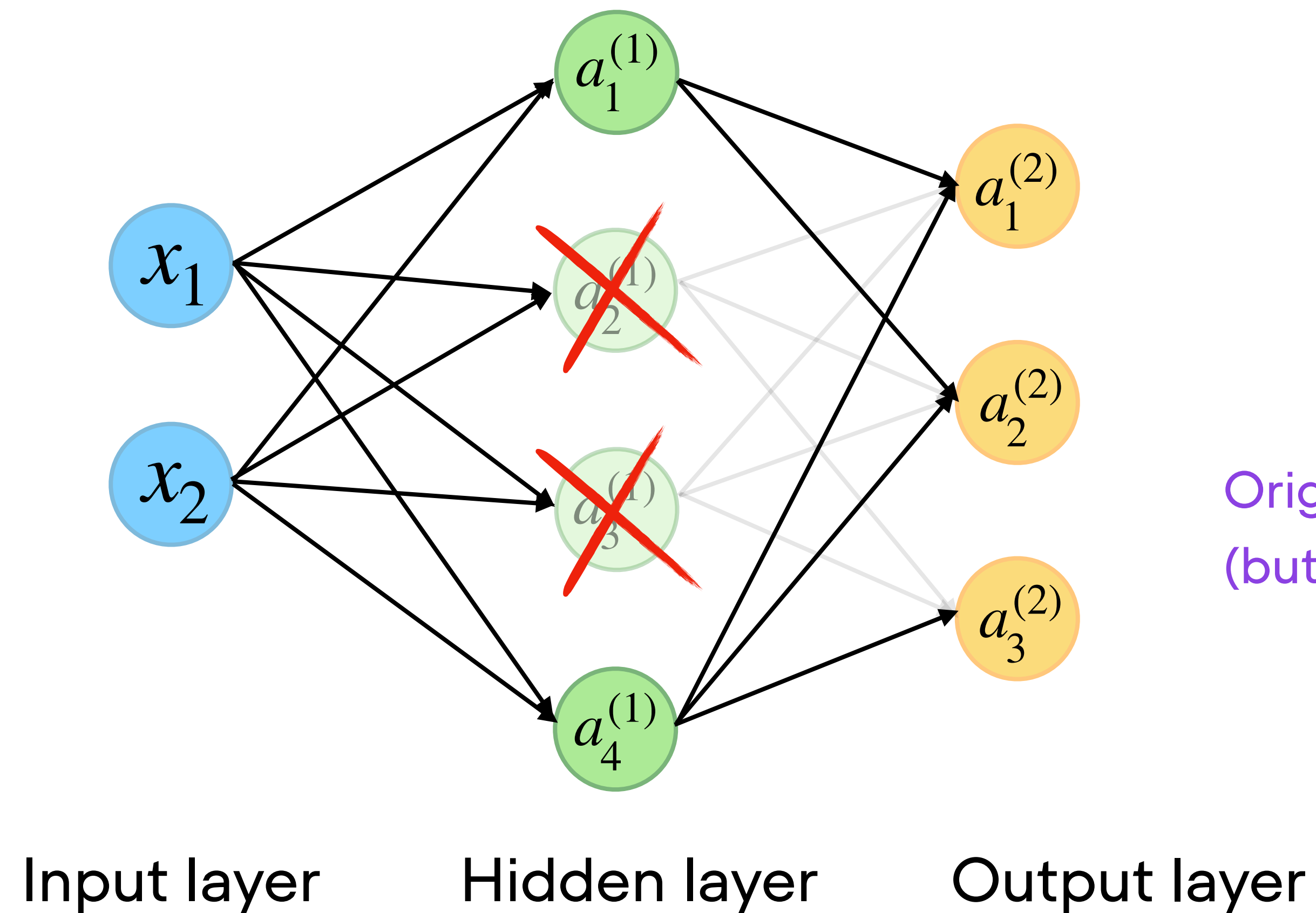


Input layer

Hidden layer

Output layer

Dropout in a nutshell: **randomly drop nodes** during training



Originally, drop probability 0.5
(but 0.2-0.8 also common now)

Why does this help against overfitting?

Network will learn not to rely on particular connections too heavily



Network will consider more connections

Next: What happens during inference?